PROBABILITY

Definition of terms

An **experiment** is an activity whose outcome is uncertain

- tossing a coin.
- investigating blood type.

A **sample** is the set of all possible outcomes of an experiment

- head or tail.
- blood type O, A, B or AB.

An **event** consists of one or more possible outcomes

- getting a head.
- blood type O.

Definition of probability

Probability is a measure of the chance of getting some particular outcome of interest from some trial or "experiment." For example, the experiment might be

- rolling a dice.
- performing a biopsy on breast lump tissue, or
- administering a drug to a patient having a heart attack.

Experiments produce outcomes.

- For rolling a dice, the outcomes will be 1 or 2 or 3 or 4 or 5 or 6.
- For the biopsy, the outcomes will be malignant or benign.
- For the patient with a heart attack, the outcomes will be to survive the first 24 hours or to die within 24 hours.

The outcomes of interest may be

- the dice giving a 6,
- the patient surviving,
- the biopsy being benign, and so on.

Basic ideas in probability

- The probability of getting any particular outcome will always lie between zero and one. The smaller the probability is, the lesser is the chance of the outcome and vice versa.
- The probability of an outcome that is certain to happen is equal to one. For example, the probability that everybody dies eventually.
- The probability of an outcome that is impossible is zero. For example, throwing a seven with a normal dice.
- If an event has as much chance of happening as of not happening (like tossing a coin and getting a head), then it has a probability of $\frac{1}{2}$ or 0.5.
- If the probability of an event happening is p, then the probability of the event not happening is $1 - p$.

Calculating probability - relative frequency

The probability of a particular outcome is equal to the proportion of times that that outcome would occur if you were to repeat the experiment a very large number of times.
Let $p(A)$ denote the probability that outcome A occurs; then

$$p(A) = \frac{\text{Number of times outcome A occurs}}{\text{Total number of possible outcomes}}$$

Exercise 1

1) Explain the meaning of the statement - the probability of a patient with a certain illness surviving for five years is 0.75. - In the past, this is the proportion that several thousand patients with this illness have survived this long.

2) Figure 1 shows levels of satisfaction with nursing care in a hospital ward. Using proportional frequencies, if a patient is chosen at random from this group, what is the probability that the patient will be

   a) very satisfied?

   b) very dissatisfied?

Figure 1: Nursing care

Satisfaction with nursing care	Number of patients ($n = 475$)
Very satisfied	121
Satisfied	161
Neutral	90
Dissatisfied	51
Very dissatisfied	52

Types of Events

An event is an outcome of a trial or experiment. Events can be:

- **mutually exclusive events**.
  - events where the occurrence of one outcome excludes the other.
  - e.g. a patient's blood type.
  - If a patients blood type is B then he cannot have the others.
  - The four events are mutually exclusive.
    - we say blood type is either O **or** A **or** B **or** AB.
    - the sum of probabilities of all mutually exclusive events is 1.
- **independent events**.
  - events where the occurrence of one does not affect the occurrence of the other.
  - the happening of one event has no bearing on whether the other happens or not.
    - e.g. a patient's blood type **and** their HIV status

Two important rules of probability

The addition rule

The addition rule states

$$p(A \text{ or } B) = p(A) + p(B) - p(A \text{ and } B) \tag{1}$$

If two outcomes A and B are mutually exclusive then

$$p(A \text{ or } B) = p(A) + p(B) \tag{2}$$

Equation 1 gives the addition rule of probability.

Assume that in Kisumu County, the distribution of blood types is Type O = 42%, Type A = 41%, Type B = 10%, and Type AB = 5%.

a) An individual is picked at random from Kisumu County, what the probability that they have blood type O? A? B? AB?

b) What is the probability that the individual will have **either** blood group O **or** A?

   i) We know that a person cannot be in two blood groups, so these outcomes are mutually exclusive.

   ii) If two outcomes are mutually exclusive, then the addition rule of probability states that the probability of either one or the other occurring is the sum of the two individual probabilities.

   iii) Hence the probability of being in either blood group O or blood group A is

$$0.42 + 0.41 = 0.83.$$

Referring to Figure 1, what is the probability of a patient chosen at random being either satisfied OR very satisfied with their nursing care?

The multiplication rule

The multiplication rule states

$$p(\text{A and B}) = p(\text{A}) \times p(\text{B}|\text{A}) \tag{3}$$

If two outcomes A and B are independent then

$$p(\text{A and B}) = p(\text{A}) \times p(\text{B}) \tag{4}$$

Equation 3 gives the multiplication rule of probability.

Suppose you are interpreting blood type in a given county. It is known that 20% of the time a patient has blood type O in this county.

- The probability that you test one's blood and it is type O is 20%.
- The probability that a successive test is blood type O is 20%.
- This is because the results are independent of each other.

– When events are independent an outcome in no way influences the result of the next outcome.

What is the probability that two successive outcomes will both be blood type O?
Let the event that you test blood type O be A. Then

$$p(\text{A and A}) = p(A) \times p(A) = 0.2 \times 0.2 = 0.04$$

This means that if we repeat this experiment (two blood type interpretations in succession) a hundred times, on four occasions we would get two successive blood type O outcomes.

Conditional probability

Conditional probability is concerned with the probability of one outcome occurring given that another outcome has already occurred i.e. $p(A|B)$ - read probability that event A occurs given that B has occurred.

– For example, given that a baby has a birthweight less than 1500 g, what is the probability that its mother smoked while pregnant?

Conditional probabilities are calculated as:

$$p(A|B) = \frac{p(\text{A and B})}{p(B)} \tag{5}$$

A main application is in diagnostic testing and screening programs.

– What is the probability of having a disease given a positive screening test?

Conditional probability Example

Consider the following results for breast cancer screening in a county. A total of 64810 persons were screened. The result of the test was positive/negative. The persons tested either developed or did not develop breast cancer.

	Develop Breast Cancer	Breast Cancer Free	Total screened
Positive	132	983	1115
Negative	45	63650	63695
Total cancer	177	64633	64810

What is the probability of having the disease (A) given a positive screening (B) result? i.e. p(A|B)

- 1115 have positive screen and 132 of those have cancer; hence probability is $\frac{132}{1115}$.
- Using Equation(5),
  - p(cancer and positive) $= \frac{132}{64810}$
  - p(positive) $= \frac{1115}{64810}$
  - $p(A|B) = \frac{132}{64810} / \frac{1115}{64810} = \frac{132}{1115}$

Class Exercise

1. In a Local Authority (LA) area with a population of 100,000, there are 25,000 smokers, 875 of whom have experienced (non-fatal) ischaemic heart disease (IHD) in the past 5 years. Of the non-smokers, 750 have also suffered IHD in the same period. If a person is chosen at random from this LA, what is the probability that

   a) this person is a smoker?

   b) has had (non-fatal) IHD in the last 5 years?

   c) is a smoker and has suffered (non-fatal) IHD?

- The probability that a smoker is chosen is $\frac{\text{Number of smokers in population}}{\text{Total population}} = \frac{25000}{100000} = 0.25$
- The probability the person has had IHD is $\frac{875 + 750}{100000} = 0.01625$
- The probability that the person chosen is a smoker who has suffered IHD is $\frac{875}{25000} = 0.035$

Introduction

- things vary.

- the big part of statistics is describing how things vary and learning about different distributions.

- we will consider some well behaved distributions including:

  - binomial
  - Poisson
  - Normal

Probability Distributions

Definition

- It is a table or an equation that tells you the probability of each outcome when you perform some "experiment." For example:

  - Rolling a die - has six possible outcomes - 1, 2, 3, 4, 5 or 6

Outcome x	1	2	3	4	5	6
Probability of the outcome p(x)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

  - This table is the probability distribution for the dice rolling experiment.
  - The outcome of rolling the die is the random variable X.
  - *variable* - because it can take various values.
  - *random* - because it is an outcome of an experiment.

- The equation of the probability distribution for this dice rolling experiment is

$$p(x) = p(X = x) = \begin{cases} \frac{1}{6}, & x = 1, 2, 3, 4, 5, 6 \\ 0 & \text{elsewhere} \end{cases}$$

Class Exercise 1

1. What is a random variable?

2. Suppose that an experiment is to toss a coin twice, write down as a table the probability distribution for the outcomes from this experiment.

Solution: Class Exercise 1

1. A random variable is a variable whose values are all the possible outcomes from a random "experiment." For example, tossing a coin, or taking a temperature, or measuring blood pressure.

2. The probability distribution is

Outcome x	HH	HT	TH	TT
Probability of the outcome p(x)	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Discrete versus continuous probability distributions

Random variables are either

- discrete (take particular values - parity of mothers in Kisumu County) or
- continuous (takes values in a given range - weight in grams of newborn babies in JOOTRH in 2020).
  - continuous variables have an unlimited or infinite number of possible values.
  - since there are an infinite number of values, the probability of any particular value has to be zero.
  - therefore we cannot show the probability distribution of a continuous random variable as a table. Instead, use an equation (called a probability density function or pdf).
  - the pdf gives the probability that the continuous random variable lies between any two defined values, for example between 2000 and 3000 g or $\leq$ 1500 g.

Hence probability distributions are either discrete or continuous. We will discuss three (3) probability distributions that arise quite often in health statistics.

- the binomial distribution.
- the Poisson distribution.
- the Normal distribution.

The Binomial distribution

The binomial experiment

A binomial experiment is an experiment which has:

- a **fixed number, n** of independent trials
    - screening $n = 100$ patients in Kisumu County for a disease
- **two possible outcomes** in each trial technically termed success (the outcome of interest) and failure.
    - they are either diseased or not diseased.
- the **probability of success in each trial is constant** and denoted p. Hence probability of failure is $1 - p$, denoted by q i.e. $q = 1 - p$.
    - it is known that in Kisumu County, 20% of the residents have the disease. Hence the probability of success $p = 20\%$.
- the binomial random variable, X denotes the number of successes in a fixed umber of trials.
    - what is the probability that you get $X = x$ successes in $n = 100$ trails?.
    - what is the probability that you get $X = 30$ diseased patients in $n = 100$ screens?.

Examples of binomial experiments

- Approximately 1 in every 20 children has a certain disease.
    - let X be the number of children with the disease out of a random sample of 100 children.
    - although the children are sampled without replacement, it is assumed that we are sampling from such a vast population that the selections are virtually independent.
    - X has a binomial distribution with $n = 100$ and $p = 1/20 = 0.05$.
- The probability of having blood type B is 0.1.
    - choose 4 people at random; X is the number with blood type B.
    - X is binomial with $n = 4$ and $p = 0.1$.

- About 10% of babies born to single parents are low birthweight babies (<2500 g).
  - there are two possible outcomes, low birthweight or not low birthweight, and the births are independent.
  - we can use the binomial distribution to tell us the probability of getting any given number of low birthweight babies.

Binomial distribution

The number of outcomes in a fixed number of trials

Binomial probability distribution - using probability rules

Consider a regular deck of 52 cards, 13 cards of each suit - hearts, diamonds, clubs and spades. Select 3 cards at random with replacement.

Let X be the number of diamond cards you get out of 3.

The experiment is a Binomial experiment because

- a fixed number, 3, of trials - n = 3.
- sampling with replacement - so trials are independent.
- there are two possible outcomes - select a diamond (D)/do not select a diamond (N).
- p = p(success) = p(select a diamond) = $\frac{1}{4}$

We can find the probability of each outcome:

Outcome	X	Probability
NNN	0	$_4 \times _4 \times _4 = \; _4$
NND	1	$_4 \times _4 \times _4 = \; _4 \; \frac{3}{4} \; ^2$
NDN	1	$_4 \times _4 \times _4 = \; _4 \; \frac{3}{4} \; ^2$
NDD	2	$_4 \times _4 \times _4 = \; _4 \; ^2 \; \frac{3}{4} \; ^2$
DNN	1	$_4 \times _4 \times _4 = \; _4 \; \frac{3}{4} \; ^2$
DND	2	$_4 \times _4 \times _4 = \; _4 \; ^2 \; \frac{3}{4} \; ^2$
DND	2	$_4 \times _4 \times _4 = \; _4 \; ^2 \; \frac{3}{4} \; ^2$
DDD	3	$_4 \times _4 \times _4 = \; _4$

Using the addition rule, we can condense the information into a table to construct the actual probability distribution.

X	0	1	2	3
$p(x) = p(X = x)$	$1\left(\frac{3}{4}\right)^{3}$	$3\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^{2}$	$3\left(\frac{1}{4}\right)^{2}\left(\frac{3}{4}\right)^{2}$	$1\left(\frac{1}{4}\right)^{3}$

In order to establish a general formula for the probability that a binomial random variable X takes any given value x, we look for patterns in the above distribution. From the way we constructed the probability distribution, we know that in general:

$p(X = x)$ = (Number of possible outcomes with x successes out of 3) $\times$ (Probability of each of the outcomes that has x successes out of 3).

1. Consider the probability that there are x successes out of 3 where $p(\text{success} = \frac{1}{4})$

   – probability of x successes and $3 - x$ failures is $(\frac{1}{4})^{x} \times (\frac{3}{4})^{3-x}$

2. Number of ways one can get x successes out of 3 trails (is harder and so we give a formula)

$$\binom{3}{x} = \frac{3!}{x!(3-x)!}$$

$$\therefore\ p(X = x) = \frac{3!}{x!(3-x)!}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{3-x}$$

In general, the probability formula for $p(X = x)$ is

$$p(X = x) = \frac{n!}{x!(n-x)!}p^{x}(1-p)^{n-x} \qquad (6)$$

where x may take values 0, 1, 2, ..., n.

Example 1

A physician is investigating the blood type of 4 patients in Maseno sub-county. It is known that 40% of the residents in Maseno have blood type A. What is the probability that 2 of the patients have blood type A?

Solution: Example 1

- What is the random variable? - the number of patients who have blood type A - could be 0, 1, 2, 3, 4.
- Is it discrete or continuous - discrete.
- What are it's properties?

- 4 blood investigations - a fixed number of trials, $n = 4$
- two possible outcomes - bloodtype A or not A
- trials are independent
- probability of success is constant in each trial, $p = 0.4$

- the random variable X is the number of successes in a fixed number of trials and has a binomial distribution.
- Hence using Equation 6,

$$p(X = 2) = \frac{4!}{2!(4-2)!} 0.4^2 (1 - 0.4)^{4-2} = 0.3456$$

Defining a binomial distribution

- To define a binomial random variable, we state the number of independent trials, n and the probability of success, p.
- n and p are called the parameters of the binomial distribution.
- We write $X \sim \text{Bin}(n, p)$ read X is distributed as binomial with parameters n and p.

Mean and variance of a binomial distribution

Example 2

Overall the proportion of people with blood type B is 0.1, i.e. 10% of the population has blood type B.

Suppose we sample 120 people at random. On average, how many would you expect to have blood type B?

Answer: 12 i.e 120×0.1.

Hence, if X is binomial with parameters n and p, then the mean or **expected value** of X is

$$\mu_X = np$$

If X is binomial with parameters n and p, then the variance and SD (**standard deviation**) of X are

$$\begin{aligned} \sigma_X^2 &= np(1-p) \\ \sigma_X &= \sqrt{np(1-p)} \end{aligned}$$

Suppose we sample 120 people at random. The number with blood type B should be about 12, give or take how many?

In other words, what is the standard deviation of the number X who have blood type B?

Answer:

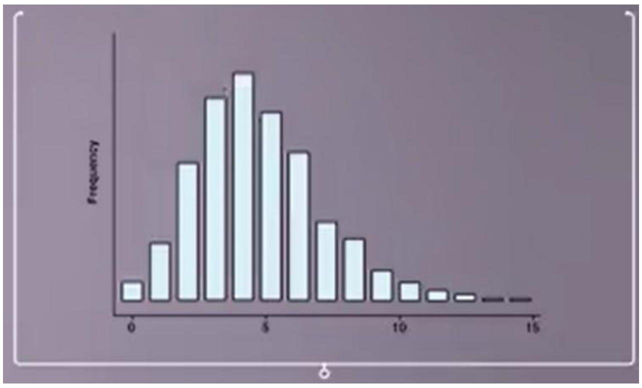
$$\sigma_X^2 = 120 \times 0.1 \times (1 - 0.1) = 10.8$$
$$\sigma_X = \sqrt{10.8} \approx 3.3$$

In a random sample of 120 people, we should expect there are about 12 with blood type B give or take about 3.3.

The Poisson distribution

- the Poisson probability distribution describes the outcome of a discrete random variable.
  - particularly count variables.
- it describes the probability of a given number of events occurring in a fixed interval of time or space.
- For example
  - the number of stillbirths in a particular hospital in a given year.
  - the number of deaths in a year.
  - the number of arrivals at an emergency department in a given 24-hour period.
    - if the common number is 4, yet sometimes 0 or 3 or 7 and so on may occur, the distribution can be represented as in Figure 2.

Figure 2: Poisson distribution with mean 4



    - we say X has a Poisson distribution with parameter 4, denoted $X \sim P_o(4)$.
    - the parameter of the Poisson distribution denotes the average (or expected) number of events in the fixed interval of time or space.
  - In general we say, $X \sim P_o(\lambda)$
  - the mean and variance of a Poisson distribution are equal and equal to λ.
    - we only need the mean, λ, to define the Poisson distribution.

The formula for the Poisson distribution is

$$p(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

where x can take values 0, 1, 2, 3, ...

However, we can use Poisson probability calculators.

Conditions for a Poisson distribution

If X is to have a Poisson distribution then the events must occur:

- singly in space or time (no clustering).
  - multiple accident victims sent to the casualty department will violate this assumption.
- independently of each other.
- at a constant rate in the sense that the mean number of occurrences in the interval is proportional to the length of the interval.

Example 3

Suppose we know that the average number of stillbirths in a year in a certain hospital maternity unit averages 10 in any 12-week period. What is the probability of getting eight stillbirths in the next 12 weeks?

Answer: Using a Poisson calculator, 0.113.

Reference

1) David Bowers (2020) Medical Statistics from Scratch: An Introduction for Health Professionals. Fourth Edition. John Wiley and Sons Ltd.

BINOMIAL AND POISSON DISTRIBUTION

BINOMIAL DISTRIBUTION

A Bernoulli trial refers to an individual event or experiment that has two possible outcomes, usually referred to as success and failure.

A binomial distribution is a probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials. It is characterized by two parameters: the number of trials (n) and the probability of success in each trial (p). The binomial distribution calculates the probabilities of observing different numbers of successes (ranging from 0 to n) based on these parameters.

Here are examples of Bernoulli trails:

1. **Public Health: Vaccination Success Rate**
   In a public health study, the Bernoulli trial could be the success rate of a specific vaccine in preventing a particular disease. Let us consider a trial where the vaccination status (success or failure) of individuals against a disease is recorded. The number of trials can be the total number of individuals in the study. The two possible outcomes are success (individuals who are successfully vaccinated) and failure (individuals who are not vaccinated or do not respond to the vaccine). The probability of success would be the efficacy rate of the vaccine. Assuming the trials are independent, the binomial random variable would represent the count of individuals who have successfully responded to the vaccine out of the total number of individuals in the study.

2. **Biomedical Sciences: Drug Efficacy in Clinical Trials**
   In a biomedical study, the Bernoulli trial could be the efficacy of a drug in treating a specific condition. Let us consider a clinical trial where the drug's effectiveness is assessed. The number of trials can be the total number of participants in the clinical trial. The two possible outcomes are success (participants who respond positively to the drug) and failure (participants who do not respond or experience adverse effects). The probability of success would be the proportion of participants who positively respond to the drug. Assuming the trials are independent, the binomial random variable would represent the count of participants who respond positively to the drug out of the total number of participants in the trial.

3. **Biology: Germination Success in Plant Seeds**
   In a biology study, the Bernoulli trial could be the germination success of plant seeds under specific conditions. Let us consider an experiment where plant seeds are subjected to different treatments, and the germination status is observed. The number of trials can be the total number of plant seeds used in the experiment. The two possible outcomes are success (seeds that successfully germinate) and failure (seeds that do not germinate). The probability of success would be the likelihood of a seed germinating under the given conditions. Assuming the trials are independent, the binomial random variable would represent the count of seeds that successfully germinate out of the total number of seeds in the experiment.

The probability function of a binomial distribution is given by:

$$p(XX = x) = \binom{n}{x} p^x(1-p)^{n-x}, x = 0, 1, \dots, n \text{ and } XX \sim Bin(n, p)$$

The mean of the number of successes is $\mu = np$ and the variance $\sigma^2 = np(1-p)$.

POISSON DISTRIBUTION

The Poisson distribution is a probability distribution that models the number of events that occur in a fixed interval of time or space, given that these events occur at a constant average rate and are independent of each other. It is often used to describe rare events or phenomena that happen randomly over a continuous interval.

To illustrate the concept of the Poisson distribution, consider three examples relevant to the fields of public health, biomedical sciences, and biology:

1. **Public Health: Emergency Room Visits**
   In a public health study, the Poisson distribution can be used to model the number of emergency room visits per hour in a hospital. The average rate of visits per hour can be estimated based on historical data. The Poisson distribution would then provide the probabilities of observing different numbers of visits within a specific hour, considering that the visits occur randomly and are independent of each other.
2. **Biomedical Sciences: Cell Mutation Rates**
   In biomedical research, the Poisson distribution can be applied to study the occurrence of cell mutations. For example, researchers might investigate the number of mutations that arise in a specific gene per unit of time. If mutations occur randomly and independently within the gene, the Poisson distribution can be used to calculate the probabilities of observing different mutation counts in a given time frame.
3. **Biology: Bacterial Colony Growth**
   In the field of biology, the Poisson distribution can be used to model the growth of bacterial colonies. Researchers may be interested in studying the number of bacterial colonies that form in a petri dish over a specific period. Given that colony formation can be considered a rare event occurring randomly and independently, the Poisson distribution can be employed to estimate the probabilities of observing different colony counts.

The Poisson distribution allows us to calculate the probabilities of observing different numbers of events (e.g., emergency room visits, cell mutations, bacterial colonies) within a fixed interval, based on the average rate of occurrence. It provides a useful tool for understanding and analyzing rare and random events in various fields of study.

The probability function of a Poisson distribution is given by:

$$p(XX = x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \dots \text{and } XX \sim P_o(\lambda)$$

The mean of the number of occurrences in a fixed interval is $\mu = \lambda$ and the variance $\sigma^2 = \lambda$.

DIFFERENCE BETWEEN THE TWO DISCRETE DISTRIBUTIONS

1. **Proportions and the Binomial Distribution**:
   Proportions are often associated with the binomial distribution. The binomial distribution models the number of successes (or "positive outcomes") in a fixed number of independent Bernoulli trials, where each trial has two possible outcomes: success (with probability p) or failure (with probability $q = 1 - p$). The proportion, in this case, represents the ratio of successes to the total number of trials.
   For example, in a public health study, you may be interested in estimating the proportion of individuals within a population who have a specific health condition. You can conduct a survey or sample a subset of the population and classify individuals as either having the condition (success) or not having it

(failure). The proportion of individuals with the condition can be calculated by dividing the number of successes by the total number of trials (sample size).

The binomial distribution allows you to calculate the probabilities of different proportions (or proportions within a certain range) based on the number of trials and the probability of success for each trial. It focuses on discrete outcomes and provides a framework for analyzing proportions in a binary setting.

2. **Rates and the Poisson Distribution**:

   Rates are commonly associated with the Poisson distribution. The Poisson distribution models the number of events that occur in a fixed interval of time or space, given a constant average rate of occurrence and independence between events. The rate represents the average number of events happening per unit of time or space.

   For instance, in biomedical sciences, you may be interested in studying the rate of cell mutations per unit of time. The Poisson distribution can be used to model the number of mutations that occur, assuming they happen randomly and independently at a constant average rate. The rate is the key parameter in the Poisson distribution.

   The Poisson distribution allows you to calculate the probabilities of observing different numbers of events (mutations) within a fixed interval, based on the average rate of occurrence. It focuses on the frequency or count of events and is often used to analyze rates or rare events.

In summary, proportions are associated with the binomial distribution, which models the number of successes in a fixed number of trials. Rates, on the other hand, are associated with the Poisson distribution, which models the number of events occurring in a fixed interval of time or space. The binomial distribution deals with discrete outcomes and proportions, while the Poisson distribution deals with count data and rates of occurrence.

SAMPLING DISTRIBUTION OF PROPORTIONS

In inferential statistics, the sampling distribution of proportions refers to the distribution of sample proportions that would be obtained if repeated samples of the same size were drawn from a population. It characterizes the variability in sample proportions and helps us make inferences about the population proportion.

For example, in a public health study, you may be interested in estimating the proportion of individuals within a population who have a specific health condition. By taking multiple random samples from the population, calculating the proportion of individuals with the health condition in each sample, and plotting the distribution of these sample proportions, you can obtain the sampling distribution of proportions.

The sampling distribution of proportions is often approximately normal for large sample sizes, $n \geq 30$, following the Central Limit Theorem. This allows us to use statistical techniques and confidence intervals to make inferences about the population proportion based on the sample proportion.

$$p \sim N\left(P, \frac{P(1-P)}{n}\right)$$

Understanding the sampling distribution of proportions is crucial for interpreting study results, determining the precision of estimates, and assessing the statistical significance of observed proportions. It provides a foundation for hypothesis testing, confidence interval estimation, and drawing conclusions about the population based on sample data.

BINOMIAL AND POISSON DISTRIBUTION

BINOMIAL DISTRIBUTION

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A binomial distribution is a probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials. It is characterized by two parameters: the number of trials (n) and the probability of success in each trial (p). The binomial distribution calculates the probabilities of observing different numbers of successes (ranging from 0 to n) based on these parameters.

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1. **Public Health: Vaccination Success Rate**
   In a public health study, the Bernoulli trial could be the success rate of a specific vaccine in preventing a particular disease. Let us consider a trial where the vaccination status (success or failure) of individuals against a disease is recorded. The number of trials can be the total number of individuals in the study. The two possible outcomes are success (individuals who are successfully vaccinated) and failure (individuals who are not vaccinated or do not respond to the vaccine). The probability of success would be the efficacy rate of the vaccine. Assuming the trials are independent, the binomial random variable would represent the count of individuals who have successfully responded to the vaccine out of the total number of individuals in the study.

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   In a biomedical study, the Bernoulli trial could be the efficacy of a drug in treating a specific condition. Let us consider a clinical trial where the drug's effectiveness is assessed. The number of trials can be the total number of participants in the clinical trial. The two possible outcomes are success (participants who respond positively to the drug) and failure (participants who do not respond or experience adverse effects). The probability of success would be the proportion of participants who positively respond to the drug. Assuming the trials are independent, the binomial random variable would represent the count of participants who respond positively to the drug out of the total number of participants in the trial.

3. **Biology: Germination Success in Plant Seeds**
   In a biology study, the Bernoulli trial could be the germination success of plant seeds under specific conditions. Let us consider an experiment where plant seeds are subjected to different treatments, and the germination status is observed. The number of trials can be the total number of plant seeds used in the experiment. The two possible outcomes are success (seeds that successfully germinate) and failure (seeds that do not germinate). The probability of success would be the likelihood of a seed germinating under the given conditions. Assuming the trials are independent, the binomial random variable would represent the count of seeds that successfully germinate out of the total number of seeds in the experiment.

The probability function of a binomial distribution is given by:

$$p(XX = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n \text{ and } XX \sim Bin(n, p)$$

The mean of the number of successes is $\mu = np$ and the variance $\sigma^2 = np(1-p)$.

POISSON DISTRIBUTION

The Poisson distribution is a probability distribution that models the number of events that occur in a fixed interval of time or space, given that these events occur at a constant average rate and are independent of each other. It is often used to describe rare events or phenomena that happen randomly over a continuous interval.

To illustrate the concept of the Poisson distribution, consider three examples relevant to the fields of public health, biomedical sciences, and biology:

1. **Public Health: Emergency Room Visits**
   In a public health study, the Poisson distribution can be used to model the number of emergency room visits per hour in a hospital. The average rate of visits per hour can be estimated based on historical data. The Poisson distribution would then provide the probabilities of observing different numbers of visits within a specific hour, considering that the visits occur randomly and are independent of each other.
2. **Biomedical Sciences: Cell Mutation Rates**
   In biomedical research, the Poisson distribution can be applied to study the occurrence of cell mutations. For example, researchers might investigate the number of mutations that arise in a specific gene per unit of time. If mutations occur randomly and independently within the gene, the Poisson distribution can be used to calculate the probabilities of observing different mutation counts in a given time frame.
3. **Biology: Bacterial Colony Growth**
   In the field of biology, the Poisson distribution can be used to model the growth of bacterial colonies. Researchers may be interested in studying the number of bacterial colonies that form in a petri dish over a specific period. Given that colony formation can be considered a rare event occurring randomly and independently, the Poisson distribution can be employed to estimate the probabilities of observing different colony counts.

The Poisson distribution allows us to calculate the probabilities of observing different numbers of events (e.g., emergency room visits, cell mutations, bacterial colonies) within a fixed interval, based on the average rate of occurrence. It provides a useful tool for understanding and analyzing rare and random events in various fields of study.

The probability function of a Poisson distribution is given by:

$$p(XX = x) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \ldots \text{ and } XX \sim P_o(\lambda)$$

The mean of the number of occurrences in a fixed interval is $\mu = \lambda$ and the variance $\sigma^2 = \lambda$.

DIFFERENCE BETWEEN THE TWO DISCRETE DISTRIBUTIONS

1. **Proportions and the Binomial Distribution**:
   Proportions are often associated with the binomial distribution. The binomial distribution models the number of successes (or "positive outcomes") in a fixed number of independent Bernoulli trials, where each trial has two possible outcomes: success (with probability p) or failure (with probability $q = 1 - p$). The proportion, in this case, represents the ratio of successes to the total number of trials.
   For example, in a public health study, you may be interested in estimating the proportion of individuals within a population who have a specific health condition. You can conduct a survey or sample a subset of the population and classify individuals as either having the condition (success) or not having it

(failure). The proportion of individuals with the condition can be calculated by dividing the number of successes by the total number of trials (sample size).

The binomial distribution allows you to calculate the probabilities of different proportions (or proportions within a certain range) based on the number of trials and the probability of success for each trial. It focuses on discrete outcomes and provides a framework for analyzing proportions in a binary setting.

2. **Rates and the Poisson Distribution**:

   Rates are commonly associated with the Poisson distribution. The Poisson distribution models the number of events that occur in a fixed interval of time or space, given a constant average rate of occurrence and independence between events. The rate represents the average number of events happening per unit of time or space.

   For instance, in biomedical sciences, you may be interested in studying the rate of cell mutations per unit of time. The Poisson distribution can be used to model the number of mutations that occur, assuming they happen randomly and independently at a constant average rate. The rate is the key parameter in the Poisson distribution.

   The Poisson distribution allows you to calculate the probabilities of observing different numbers of events (mutations) within a fixed interval, based on the average rate of occurrence. It focuses on the frequency or count of events and is often used to analyze rates or rare events.

In summary, proportions are associated with the binomial distribution, which models the number of successes in a fixed number of trials. Rates, on the other hand, are associated with the Poisson distribution, which models the number of events occurring in a fixed interval of time or space. The binomial distribution deals with discrete outcomes and proportions, while the Poisson distribution deals with count data and rates of occurrence.

SAMPLING DISTRIBUTION OF PROPORTIONS

In inferential statistics, the sampling distribution of proportions refers to the distribution of sample proportions that would be obtained if repeated samples of the same size were drawn from a population. It characterizes the variability in sample proportions and helps us make inferences about the population proportion.

For example, in a public health study, you may be interested in estimating the proportion of individuals within a population who have a specific health condition. By taking multiple random samples from the population, calculating the proportion of individuals with the health condition in each sample, and plotting the distribution of these sample proportions, you can obtain the sampling distribution of proportions.

The sampling distribution of proportions is often approximately normal for large sample sizes, $n \geq 30$, following the Central Limit Theorem. This allows us to use statistical techniques and confidence intervals to make inferences about the population proportion based on the sample proportion.

$$p \sim N\left(P, \frac{P(1-P)}{n}\right)$$

Understanding the sampling distribution of proportions is crucial for interpreting study results, determining the precision of estimates, and assessing the statistical significance of observed proportions. It provides a foundation for hypothesis testing, confidence interval estimation, and drawing conclusions about the population based on sample data.

Module 4: Probability & Probability Distributions

Module Assignment

Source notes cover probability rules, conditional probability, random variables, binomial and Poisson distributions, with clinical examples.

1. Using a screening scenario, calculate an observed probability of a positive result.

2. Create a 2×2 clinical scenario and calculate one conditional probability, clearly identifying the numerator and denominator.

3. For a binomial scenario, define n and p and calculate the probability of a specified number of successes.

4. For a Poisson scenario involving hospital events in a fixed time interval, identify λ and calculate one requested event probability.

5. Interpret every answer in clinical language.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.